inhibitory_schur_modulation

August 2, 2026

1 Inhibitory modulation via Schur complement and perturbation analysis

This notebook investigates how inhibitory routing modulates stability in the signed, directed, weighted `mij` network.

Matrix convention used here:

- raw CSV rows are sources
- raw CSV columns are receivers
- analysis matrix is transposed to `M[receiver, source]`
- block names are written as source -> receiver: `EE`, `EI`, `IE`, `II`

Every section is shown side-by-side for two matrix variants:

- `with_self`: diagonal/self connections are retained
- `no_self`: the diagonal is zeroed before normalization

The default normalization is spectral-radius normalization to `0.85`. That means a baseline dominant eigenvalue of `0.85` is not a measured raw value; it is the chosen normalized reference point for each variant.

[1]:
```
from pathlib import Path

import numpy as np
import pandas as pd

try:
    import matplotlib.pyplot as plt
    import seaborn as sns
    sns.set_theme(style="whitegrid", context="notebook")
    HAS_PLOTS = True
except ImportError:
    HAS_PLOTS = False
    print("Plotting libraries are not installed in this environment; table␣
  ↪analyses will still run.")

from inhibitory_schur_modulation import (
    block_view,
    load_mij_data,
```

```
    run_side_by_side_analysis,
    save_side_by_side_outputs,
)

OUTPUT_DIR = Path("outputs/inhibitory_schur_modulation")
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
TARGET_RHO = 1
ALPHA = 0.85
N_NULL = 250
```

1.1 Load, orient, and normalize

The loader uses `matrices/mij_netlist.csv` to recover E/I labels, then falls back to row-sign inference only if metadata is missing. Each side of the comparison is normalized independently to the target spectral radius.

[2]:
```
data = load_mij_data("matrices/mij_matrix.csv", "matrices/mij_netlist.csv")
results = run_side_by_side_analysis(
    data,
    normalization="spectral_radius",
    target=TARGET_RHO,
    alpha=ALPHA,
    n_null=N_NULL,
)

print(f"cells: {len(data.labels)}")
print(f"excitatory: {len(data.excitatory)} | inhibitory: {len(data.
 ↪inhibitory)}")
print(f"normalization target spectral radius: {TARGET_RHO}")
results["normalization"]
```

```
cells: 85
excitatory: 32 | inhibitory: 53
normalization target spectral radius: 1
```

[2]:

	method	target	scale	spectral_radius	spectral_abscissa	\
0	spectral_radius	1	0.000038	1.0	1.0	
1	spectral_radius	1	0.000067	1.0	1.0	

	dominant_eigenvalue	dominant_real	dominant_imag	variant
0	1.0+0.0j	1.0	0.0	with_self
1	1.0+0.0j	1.0	0.0	no_self

[3]:
```
INHIBITORY_FONT_COLOR = "#c1121f"
INHIBITORY_CELLS = set(data.inhibitory)
CELLTYPE_COLUMNS = [
    "cell",
    "node",
```

```python
    "inhibitory_cell",
    "I_middle",
    "I_first",
    "I_second",
    "E_source",
    "E_receiver",
]


def is_inhibitory_celltype(value):
    return str(value) in INHIBITORY_CELLS


def style_inhibitory_celltypes(df, columns=None):
    """Render inhibitory cell-type names in red in notebook tables."""
    display_df = df.copy()
    if columns is None:
        columns = [col for col in CELLTYPE_COLUMNS if col in display_df.columns]
    else:
        columns = [col for col in columns if col in display_df.columns]

    def color_value(value):
        if is_inhibitory_celltype(value):
            return f"color: {INHIBITORY_FONT_COLOR}; font-weight: 700"
        return ""

    if not columns:
        return display_df
    styler = display_df.style
    # pandas 3.x renamed Styler.applymap() to Styler.map().
    if hasattr(styler, "map"):
        return styler.map(color_value, subset=columns)
    return styler.applymap(color_value, subset=columns)


def color_inhibitory_ticklabels(ax):
    """Color tick labels red when they are, or contain, inhibitory cell-type␣
↪names."""
    for tick in list(ax.get_xticklabels()) + list(ax.get_yticklabels()):
        text = tick.get_text()
        if text in INHIBITORY_CELLS or any(cell in text for cell in␣
↪INHIBITORY_CELLS):
            tick.set_color(INHIBITORY_FONT_COLOR)
            tick.set_fontweight("bold")
    return ax
```

[4]:
```
block_rows = []
for variant, payload in results["variants"].items():
    for name, block in block_view(payload["M"], data.ei).items():
        block_rows.append({
            "variant": variant,
            "block": name,
            "shape": str(block.shape),
            "nonzero": int(np.count_nonzero(block)),
            "positive": int(np.count_nonzero(block > 0)),
            "negative": int(np.count_nonzero(block < 0)),
            "fro_norm": np.linalg.norm(block, "fro"),
        })
pd.DataFrame(block_rows)
```

[4]:

	variant	block	shape	nonzero	positive	negative	fro_norm
0	with_self	EE	(32, 32)	382	382	0	5.171595
1	with_self	EI	(53, 32)	352	352	0	81.078433
2	with_self	IE	(32, 53)	171	0	171	0.088920
3	with_self	II	(53, 53)	736	0	736	0.423299
4	no_self	EE	(32, 32)	355	355	0	9.042052
5	no_self	EI	(53, 32)	352	352	0	143.236198
6	no_self	IE	(32, 53)	171	0	171	0.157090
7	no_self	II	(53, 53)	692	0	692	0.625358

1.2 Perturbation: remove EE, EI, IE, or II

This table compares the exact dominant eigenvalue change after zeroing each block with a first-order eigenvalue perturbation estimate.

Interpretation cue: if removing a block increases the dominant real part, that block was stabilizing relative to that variant's normalized baseline. If removing it decreases the dominant real part, that block was destabilizing or amplifying.

[5]:
```
perturbation = results["perturbation"]
perturbation.sort_values(["removed_block", "variant"])
```

[5]:

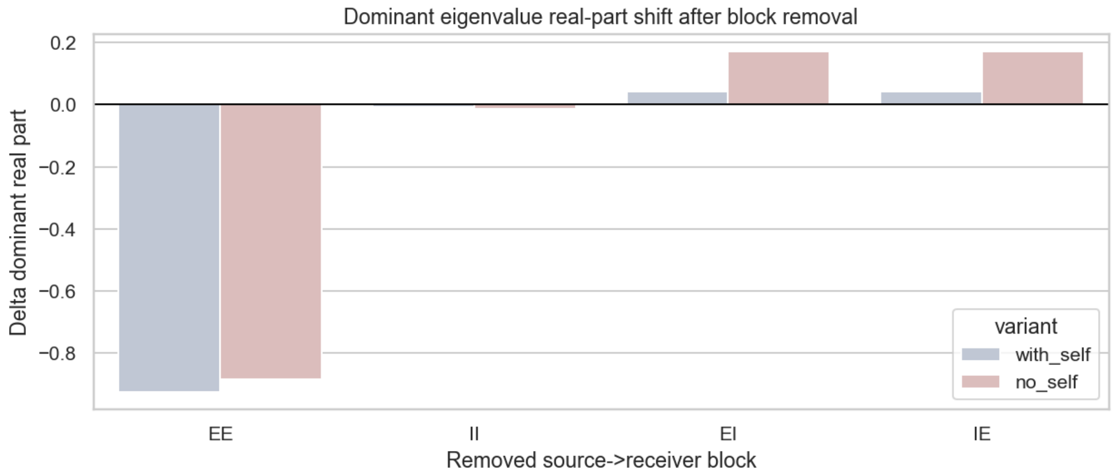
	variant	removed_block	block_meaning	dominant_real_before	\
4	no_self	EE	E source -> E receiver	1.0	
0	with_self	EE	E source -> E receiver	1.0	
7	no_self	EI	E source -> I receiver	1.0	
2	with_self	EI	E source -> I receiver	1.0	
6	no_self	IE	I source -> E receiver	1.0	
3	with_self	IE	I source -> E receiver	1.0	
5	no_self	II	I source -> I receiver	1.0	
1	with_self	II	I source -> I receiver	1.0	

	dominant_real_after	delta_dominant_real	spectral_radius_before	\
4	0.116951	-0.883049	1.0	

```
0             0.074330           -0.925670                    1.0
7             1.171643            0.171643                    1.0
2             1.041928            0.041928                    1.0
6             1.171643            0.171643                    1.0
3             1.041928            0.041928                    1.0
5             0.985519           -0.014481                    1.0
1             0.992526           -0.007474                    1.0

   spectral_radius_after  delta_spectral_radius  first_order_delta_real  \
4               0.645704              -0.354296               -1.377895
0               0.367008              -0.632992               -1.080015
7               1.171643               0.171643                0.195510
2               1.041928               0.041928                0.043133
6               1.171643               0.171643                0.195510
3               1.041928               0.041928                0.043133
5               0.985519              -0.014481               -0.013124
1               0.992526              -0.007474               -0.006251

   first_order_delta_imag
4                     0.0
0                     0.0
7                     0.0
2                     0.0
6                     0.0
3                     0.0
5                     0.0
1                     0.0
```

[6]:

```python
if HAS_PLOTS:
    fig, ax = plt.subplots(figsize=(9, 4), dpi=140)
    sns.barplot(data=perturbation, x="removed_block", y="delta_dominant_real",␣
 ↪hue="variant", ax=ax, palette="vlag")
    ax.axhline(0, color="black", linewidth=1)
    ax.set_title("Dominant eigenvalue real-part shift after block removal")
    ax.set_xlabel("Removed source->receiver block")
    ax.set_ylabel("Delta dominant real part")
    plt.tight_layout()
    fig.savefig(OUTPUT_DIR / "block_removal_delta_dominant_real_by_variant.
 ↪png", bbox_inches="tight")
    plt.show()
else:
    perturbation[["variant", "removed_block", "dominant_real_before",␣
 ↪"dominant_real_after", "delta_dominant_real"]]
```



1.3 Schur complement: eliminate inhibitory nodes

The inhibitory Schur complement is

`M_eff_E(z) = M_EE + M_EI @ inv(z I - M_II) @ M_IE`

where `z` defaults to the dominant eigenvalue of the full network variant. The feedback term is the part of E dynamics mediated by inhibitory cells, including all chained I-I paths inside the resolvent.

[7]:
```
schur_summary = results["schur_summary"]
schur_summary
```

[7]:

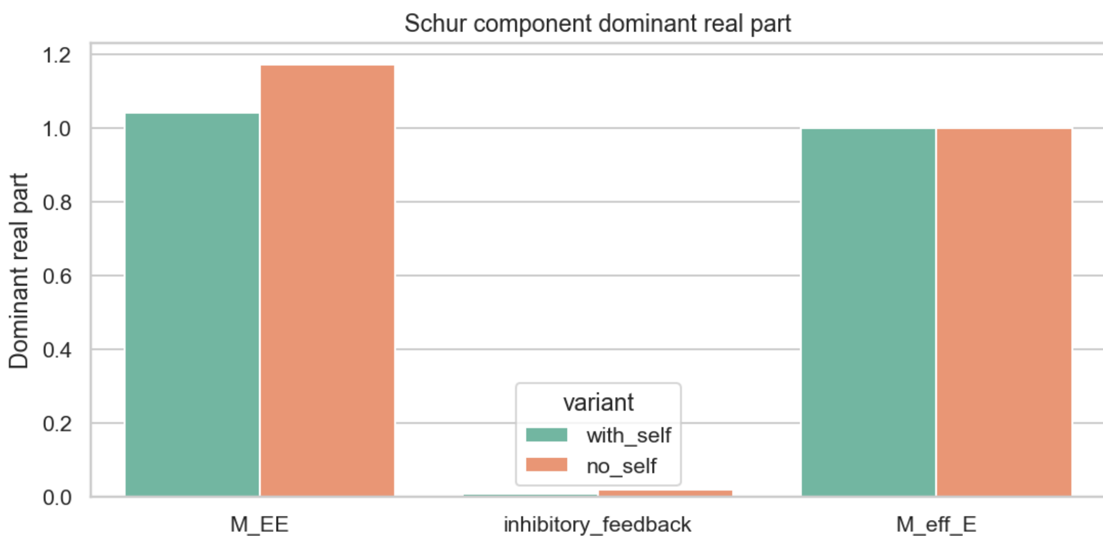
	variant	component	z_real	z_imag	rho_MII_over_z
0	with_self	M_EE	1.0	0.0	0.182206
1	with_self	inhibitory_feedback	1.0	0.0	0.182206
2	with_self	M_eff_E	1.0	0.0	0.182206
3	no_self	M_EE	1.0	0.0	0.136907
4	no_self	inhibitory_feedback	1.0	0.0	0.136907
5	no_self	M_eff_E	1.0	0.0	0.136907

	spectral_radius	spectral_abscissa	dominant_eigenvalue	dominant_real
0	1.041928	1.041928	1.041928+0.000000j	1.041928
1	0.116285	0.006103	0.006103+0.000000j	0.006103
2	1.000000	1.000000	1.000000+0.000000j	1.000000
3	1.171643	1.171643	1.171643+0.000000j	1.171643
4	0.365358	0.018307	0.018307+0.000000j	0.018307
5	1.000000	1.000000	1.000000+0.000000j	1.000000

	dominant_imag
0	0.0
1	0.0
2	0.0

```
3              0.0
4              0.0
5              0.0
```

[8]:
```
if HAS_PLOTS:
    fig, ax = plt.subplots(figsize=(8, 4), dpi=140)
    sns.barplot(data=schur_summary, x="component", y="dominant_real",␣
 ↪hue="variant", ax=ax, palette="Set2")
    ax.axhline(0, color="black", linewidth=1)
    ax.set_title("Schur component dominant real part")
    ax.set_xlabel("")
    ax.set_ylabel("Dominant real part")
    plt.tight_layout()
    fig.savefig(OUTPUT_DIR / "schur_summary_by_variant.png",␣
 ↪bbox_inches="tight")
    plt.show()
else:
    schur_summary[["variant", "component", "dominant_real", "spectral_radius",␣
 ↪"rho_MII_over_z"]]
```



1.4 Enrichment: excite all inhibitory or all excitatory cells

The response uses the stable discrete resolvent `(I - alpha M)^-1 u`. Positive E responses after inhibitory excitation are candidate disinhibitory effects; large I responses identify inhibitory relays.

[9]:
```
print("Top response to exciting all inhibitory cells")
display(style_inhibitory_celltypes(results["response_I"]))
print("Top response to exciting all excitatory cells")
display(style_inhibitory_celltypes(results["response_E"]))
```

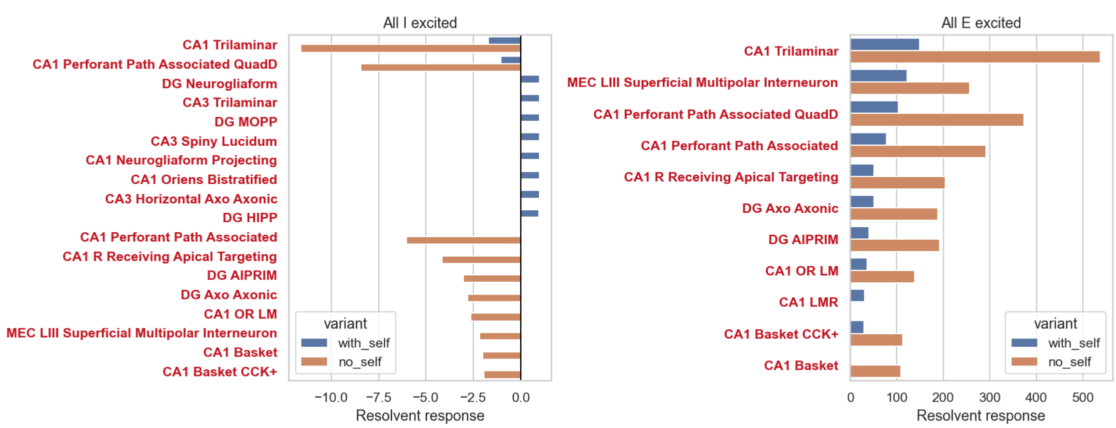
```
Top response to exciting all inhibitory cells
```

```
<pandas.io.formats.style.Styler at 0x220dce27790>
```

```
Top response to exciting all excitatory cells
```

```
<pandas.io.formats.style.Styler at 0x220dce00bd0>
```

[10]:
```
if HAS_PLOTS:
    fig, axes = plt.subplots(1, 2, figsize=(13, 5), dpi=140)
    for ax, (title, frame) in zip(axes, [("All I excited",␣
 ↪results["response_I"]), ("All E excited", results["response_E"])]):
        plot_frame = frame.groupby("variant", group_keys=False).head(10)
        sns.barplot(data=plot_frame, y="cell", x="response", hue="variant",␣
 ↪ax=ax)
        ax.axvline(0, color="black", linewidth=1)
        ax.set_title(title)
        ax.set_xlabel("Resolvent response")
        ax.set_ylabel("")
        color_inhibitory_ticklabels(ax)
    plt.tight_layout()
    fig.savefig(OUTPUT_DIR / "population_excitation_responses_by_variant.png",␣
 ↪bbox_inches="tight")
    plt.show()
else:
    display(style_inhibitory_celltypes(pd.concat([results["response_I"].
 ↪head(10), results["response_E"].head(10)], ignore_index=True)))
```



1.5 Motif enrichment and stability effect

The motif table now includes all-excitatory and all-inhibitory motif classes, along with convergent, divergent, chain, reciprocal, E-I-E, and E-I-I-E motifs.

The companion stability table asks the causal perturbation question: if we remove the edges participating in a motif class, how does the dominant eigenvalue change? This mirrors the logic of sections 7.1-7.4 in the reference notebook: define a condition, compute the network response, and summarize E/I population-level stability effects.

[11]:
```
motif = results["motif"]
motif_stability = results["motif_stability"]

print("Motif enrichment/counts")
display(motif.sort_values(["motif", "variant"]))

print("Motif-removal dominant eigenvalue stability effects")
display(motif_stability.sort_values(["motif", "variant"])[[
    "variant",
    "motif",
    "motif_count",
    "edges_removed",
    "dominant_real_before",
    "dominant_real_after",
    "delta_dominant_real",
    "spectral_radius_before",
    "spectral_radius_after",
    "delta_spectral_radius",
]])
```

```
Motif enrichment/counts

      variant                              motif  count  null_mean  \
10    no_self             E_to_I_to_I_to_E_loop  11255   1520.248
0   with_self             E_to_I_to_I_to_E_loop  12468   1701.480
16    no_self          all_excitatory_E_E_E_chains   4472   4472.000
6   with_self          all_excitatory_E_E_E_chains   4472   4472.000
19    no_self          all_excitatory_edges_E_to_E    355    355.000
9   with_self          all_excitatory_edges_E_to_E    355    355.000
15    no_self          all_inhibitory_I_I_I_chains   9553   9553.000
5   with_self          all_inhibitory_I_I_I_chains   9553   9553.000
17    no_self          all_inhibitory_edges_I_to_I    692    692.000
7   with_self          all_inhibitory_edges_I_to_I    692    692.000
14    no_self   convergent_two_sources_one_receiver  15879  15879.000
4   with_self   convergent_two_sources_one_receiver  15879  15879.000
12    no_self               directed_two_step_chain  28037  28037.000
2   with_self               directed_two_step_chain  28037  28037.000
11    no_self              disinhibitory_I_to_I_to_E   1805    541.652
1   with_self              disinhibitory_I_to_I_to_E   1956    593.712
13    no_self    divergent_one_source_two_receivers  19285  19285.000
3   with_self    divergent_one_source_two_receivers  19285  19285.000
18    no_self                       reciprocal_pair    422    422.000
8   with_self                       reciprocal_pair    422    422.000

        null_sd    z_score  empirical_p_ge
10   179.012156  54.380396        0.003984
```

0	191.449726	56.236800	0.003984
16	NaN	NaN	1.000000
6	NaN	NaN	1.000000
19	NaN	NaN	1.000000
9	NaN	NaN	1.000000
15	NaN	NaN	1.000000
5	NaN	NaN	1.000000
17	NaN	NaN	1.000000
7	NaN	NaN	1.000000
14	NaN	NaN	1.000000
4	NaN	NaN	1.000000
12	NaN	NaN	1.000000
2	NaN	NaN	1.000000
11	43.308584	29.170845	0.003984
1	46.345882	29.393938	0.003984
13	NaN	NaN	1.000000
3	NaN	NaN	1.000000
18	NaN	NaN	1.000000
8	NaN	NaN	1.000000

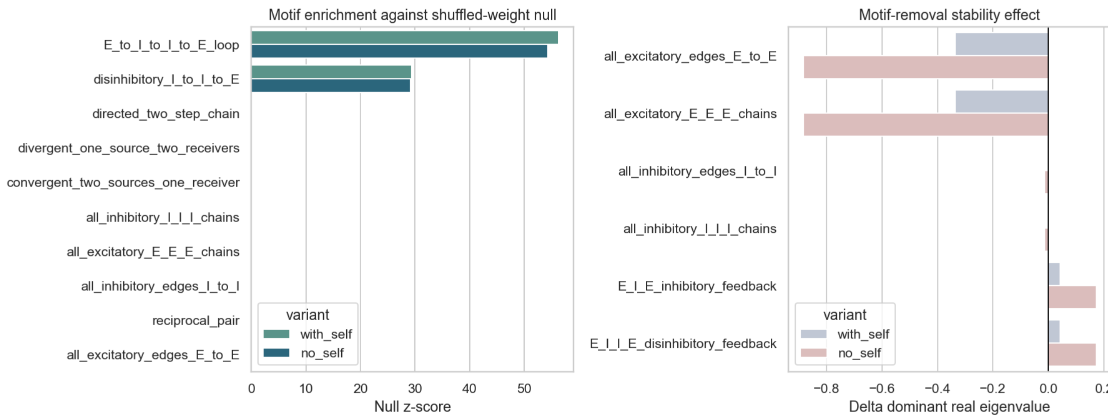
Motif-removal dominant eigenvalue stability effects

	variant	motif	motif_count	edges_removed	\
11	no_self	E_I_E_inhibitory_feedback	1341	515	
4	with_self	E_I_E_inhibitory_feedback	1341	515	
10	no_self	E_I_I_E_disinhibitory_feedback	11255	1159	
5	with_self	E_I_I_E_disinhibitory_feedback	12468	1202	
7	no_self	all_excitatory_E_E_E_chains	4472	355	
1	with_self	all_excitatory_E_E_E_chains	4472	355	
6	no_self	all_excitatory_edges_E_to_E	355	355	
0	with_self	all_excitatory_edges_E_to_E	355	355	
9	no_self	all_inhibitory_I_I_I_chains	9553	692	
3	with_self	all_inhibitory_I_I_I_chains	9553	692	
8	no_self	all_inhibitory_edges_I_to_I	692	692	
2	with_self	all_inhibitory_edges_I_to_I	692	692	

	dominant_real_before	dominant_real_after	delta_dominant_real	\
11	1.0	1.171643	0.171643	
4	1.0	1.041928	0.041928	
10	1.0	1.171643	0.171643	
5	1.0	1.041928	0.041928	
7	1.0	0.116951	-0.883049	
1	1.0	0.662927	-0.337073	
6	1.0	0.116951	-0.883049	
0	1.0	0.662927	-0.337073	
9	1.0	0.985519	-0.014481	
3	1.0	0.999186	-0.000814	
8	1.0	0.985519	-0.014481	
2	1.0	0.999186	-0.000814	

	spectral_radius_before	spectral_radius_after	delta_spectral_radius
11	1.0	1.171643	0.171643
4	1.0	1.041928	0.041928
10	1.0	1.171643	0.171643
5	1.0	1.041928	0.041928
7	1.0	0.645704	-0.354296
1	1.0	0.662927	-0.337073
6	1.0	0.645704	-0.354296
0	1.0	0.662927	-0.337073
9	1.0	0.985519	-0.014481
3	1.0	0.999186	-0.000814
8	1.0	0.985519	-0.014481
2	1.0	0.999186	-0.000814

[12]:
```python
if HAS_PLOTS:
    fig, axes = plt.subplots(1, 2, figsize=(13, 5), dpi=140)
    sns.barplot(data=motif, y="motif", x="z_score", hue="variant", ax=axes[0],␣
↪palette="crest")
    axes[0].axvline(0, color="black", linewidth=1)
    axes[0].set_title("Motif enrichment against shuffled-weight null")
    axes[0].set_xlabel("Null z-score")
    axes[0].set_ylabel("")

    sns.barplot(data=motif_stability, y="motif", x="delta_dominant_real",␣
↪hue="variant", ax=axes[1], palette="vlag")
    axes[1].axvline(0, color="black", linewidth=1)
    axes[1].set_title("Motif-removal stability effect")
    axes[1].set_xlabel("Delta dominant real eigenvalue")
    axes[1].set_ylabel("")
    plt.tight_layout()
    fig.savefig(OUTPUT_DIR / "motif_enrichment_and_stability_by_variant.png",␣
↪bbox_inches="tight")
    plt.show()
else:
    motif_stability[["variant", "motif", "motif_count", "delta_dominant_real",␣
↪"delta_spectral_radius"]]
```



1.6 Neumann series for chained I-I effects

The Schur inhibitory feedback can be expanded as a Neumann series. Order `k` is the number of chained I-I edges inside the inhibitory block:

- `k = 0`: `E->I->E`
- `k = 1`: `E->I->I->E`
- `k = 2`: `E->I->I->I->E`
- `k = 3`: `E->I->I->I->I->E`

Model-selection rules:

- contribution decay: keep orders whose relative contribution is still large
- cumulative energy: keep enough orders to explain the requested cumulative energy
- spectral radius: stop when the spectral tail bound is small
- null significance: keep orders above the shuffled-II null model

[13]:
```
neumann = results["neumann"]
print("Selected order by variant")
display(neumann["selected_orders"])
print("Selected order by rule")
display(neumann["selected_by_rule"])
neumann["selection_table"]
```

```
Selected order by variant
```

	variant	selected_order	alpha_rho_MII
0	with_self	2	0.154875
1	no_self	1	0.116371

```
Selected order by rule
```

	variant	rule	selected_order
0	with_self	contribution_decay	1
1	with_self	cumulative_energy	0
2	with_self	spectral_radius	2

3	with_self	null_significance	12
4	no_self	contribution_decay	1
5	no_self	cumulative_energy	0
6	no_self	spectral_radius	1
7	no_self	null_significance	0

[13]:

	variant	ii_chain_order	motif_length \
0	with_self	0	E->I->E
1	with_self	1	E->I->I->E
2	with_self	2	E->I->I->I->E
3	with_self	3	E->I->I->I->I->E
4	with_self	4	E->I->I->I->I->I->E
5	with_self	5	E->I->I->I->I->I->I->E
6	with_self	6	E->I->I->I->I->I->I->I->E
7	with_self	7	E->I->I->I->I->I->I->I->I->E
8	with_self	8	E->I->I->I->I->I->I->I->I->I->E
9	with_self	9	E->I->I->I->I->I->I->I->I->I->I->E
10	with_self	10	E->I->I->I->I->I->I->I->I->I->I->I->E
11	with_self	11	E->I->I->I->I->I->I->I->I->I->I->I->I->E
12	with_self	12	E->I->I->I->I->I->I->I->I->I->I->I->I->I->E
13	no_self	0	E->I->E
14	no_self	1	E->I->I->E
15	no_self	2	E->I->I->I->E
16	no_self	3	E->I->I->I->I->E
17	no_self	4	E->I->I->I->I->I->E
18	no_self	5	E->I->I->I->I->I->I->E
19	no_self	6	E->I->I->I->I->I->I->I->E
20	no_self	7	E->I->I->I->I->I->I->I->I->E
21	no_self	8	E->I->I->I->I->I->I->I->I->I->E
22	no_self	9	E->I->I->I->I->I->I->I->I->I->I->E
23	no_self	10	E->I->I->I->I->I->I->I->I->I->I->I->E
24	no_self	11	E->I->I->I->I->I->I->I->I->I->I->I->I->E
25	no_self	12	E->I->I->I->I->I->I->I->I->I->I->I->I->I->E

	fro_norm	signed_sum	positive_sum	negative_sum	max_abs \
0	2.234204e+00	-5.574129e+00	0.000000e+00	-5.574129e+00	2.135653e+00
1	9.093412e-02	4.585259e-01	4.585259e-01	0.000000e+00	4.456406e-02
2	1.060087e-02	-5.372051e-02	0.000000e+00	-5.372051e-02	5.875143e-03
3	1.536676e-03	7.175063e-03	7.175063e-03	0.000000e+00	9.099114e-04
4	2.325226e-04	-1.011259e-03	0.000000e+00	-1.011259e-03	1.409223e-04
5	3.568547e-05	1.469741e-04	1.469741e-04	0.000000e+00	2.182531e-05
6	5.506304e-06	-2.180583e-05	0.000000e+00	-2.180583e-05	3.380190e-06
7	8.514547e-07	3.281165e-06	3.281165e-06	0.000000e+00	5.235061e-07
8	1.317793e-07	-4.984576e-07	0.000000e+00	-4.984576e-07	8.107787e-08
9	2.040311e-08	7.620921e-08	7.620921e-08	0.000000e+00	1.255692e-08
10	3.159493e-09	-1.170148e-08	0.000000e+00	-1.170148e-08	1.944749e-09
11	4.892948e-10	1.801813e-09	1.801813e-09	0.000000e+00	3.011926e-10

```
12  7.577718e-11 -2.779729e-10  0.000000e+00 -2.779729e-10  4.664713e-11
13  6.972970e+00 -1.739690e+01  0.000000e+00 -1.739690e+01  6.665389e+00
14  2.530079e-01  9.184770e-01  9.184770e-01  0.000000e+00  1.418198e-01
15  2.419399e-02 -9.178299e-02  0.000000e+00 -9.178299e-02  1.450280e-02
16  2.698699e-03  9.456404e-03  9.456404e-03  0.000000e+00  1.670668e-03
17  3.136776e-04 -1.072519e-03  0.000000e+00 -1.072519e-03  1.950827e-04
18  3.640330e-05  1.212799e-04  1.212799e-04  0.000000e+00  2.267000e-05
19  4.232650e-06 -1.387737e-05  0.000000e+00 -1.387737e-05  2.639380e-06
20  4.921994e-07  1.592311e-06  1.592311e-06  0.000000e+00  3.070638e-07
21  5.726977e-08 -1.833826e-07  0.000000e+00 -1.833826e-07  3.573489e-08
22  6.664619e-09  2.116865e-08  2.116865e-08  0.000000e+00  4.158183e-09
23  7.757036e-10 -2.448155e-09  0.000000e+00 -2.448155e-09  4.838840e-10
24  9.028825e-11  2.835430e-10  2.835430e-10  0.000000e+00  5.630813e-11
25  1.050939e-11 -3.287482e-11  0.000000e+00 -3.287482e-11  6.552505e-12

    spectral_bound  relative_contribution  cumulative_energy  \
0     1.000000e+00           9.557886e-01           0.955789
1     1.548748e-01           3.890145e-02           0.994690
2     2.398619e-02           4.535034e-03           0.999225
3     3.714856e-03           6.573871e-04           0.999882
4     5.753374e-04           9.947274e-05           0.999982
5     8.910524e-05           1.526618e-05           0.999997
6     1.380015e-05           2.355587e-06           1.000000
7     2.137295e-06           3.642508e-07           1.000000
8     3.310131e-07           5.637494e-08           1.000000
9     5.126558e-08           8.728415e-09           1.000000
10    7.939744e-09           1.351625e-09           1.000000
11    1.229666e-09           2.093194e-10           1.000000
12    1.904442e-10           3.241734e-11           1.000000
13    1.000000e+00           9.613613e-01           0.961361
14    1.163708e-01           3.488213e-02           0.996243
15    1.354217e-02           3.335618e-03           0.999579
16    1.575914e-03           3.720688e-04           0.999951
17    1.833905e-04           4.324664e-05           0.999994
18    2.134130e-05           5.018913e-06           0.999999
19    2.483506e-06           5.835542e-07           1.000000
20    2.890076e-07           6.785939e-08           1.000000
21    3.363206e-08           7.895766e-09           1.000000
22    3.913792e-09           9.188491e-10           1.000000
23    4.554512e-10           1.069460e-10           1.000000
24    5.300124e-11           1.244801e-11           1.000000
25    6.167800e-12           1.448926e-12           1.000000

    passes_decay_rule  passes_cumulative_energy_rule  \
0                True                          False
1                True                          False
2               False                          False
```

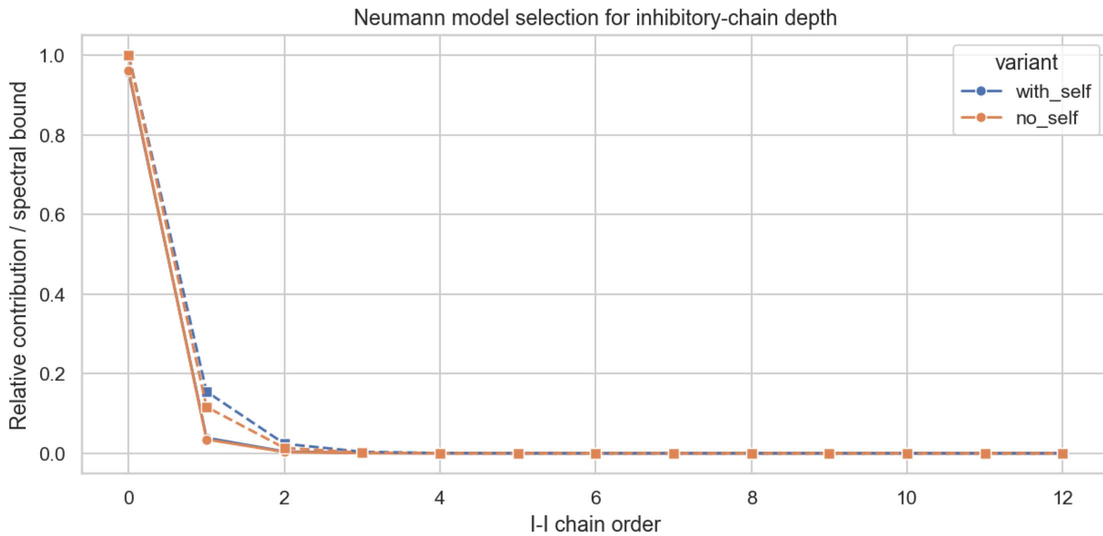
```
3            False                        False
4            False                        False
5            False                        False
6            False                        False
7            False                        False
8            False                        False
9            False                        False
10           False                        False
11           False                        False
12           False                        False
13            True                        False
14            True                        False
15           False                        False
16           False                        False
17           False                        False
18           False                        False
19           False                        False
20           False                        False
21           False                        False
22           False                        False
23           False                        False
24           False                        False
25           False                        False

    passes_spectral_radius_rule  null_mean_fro   null_sd_fro     null_z  \
0                          True   2.234204e+00  1.067952e-14   0.997998
1                          True   8.559056e-02  5.132055e-02   0.104121
2                          True   6.540813e-03  4.643704e-03   0.874315
3                         False   5.482822e-04  4.444457e-04   2.223879
4                         False   4.888299e-05  5.297025e-05   3.466844
5                         False   4.474300e-06  6.373738e-06   4.896840
6                         False   4.269080e-07  7.778713e-07   6.529867
7                         False   4.244724e-08  9.525951e-08   8.492668
8                         False   4.389466e-09  1.173299e-08  10.857406
9                         False   4.717459e-10  1.453215e-09  13.715363
10                        False   5.259686e-11  1.813980e-10  17.127509
11                        False   6.083583e-12  2.287035e-11  21.128282
12                        False   7.270466e-13  2.920924e-12  25.693965
13                         True   6.972970e+00  2.402892e-14  -0.997998
14                         True   3.849930e-01  2.451001e-01  -0.538494
15                        False   4.733520e-02  3.316057e-02  -0.697853
16                        False   5.864996e-03  3.374477e-03  -0.938308
17                        False   7.569858e-04  5.673323e-04  -0.781391
18                        False   9.865779e-05  7.545348e-05  -0.825071
19                        False   1.330623e-05  1.348374e-05  -0.672927
20                        False   1.781609e-06  1.869655e-06  -0.689651
21                        False   2.457306e-07  3.277723e-07  -0.574975
```

```
22                      False   3.412668e-08  4.840638e-08  -0.567323
23                      False   4.787465e-09  8.195014e-09  -0.489537
24                      False   6.896312e-10  1.297661e-09  -0.461864
25                      False   9.861128e-11  2.111242e-10  -0.417299

    passes_null_significance_rule
0                           False
1                           False
2                           False
3                            True
4                            True
5                            True
6                            True
7                            True
8                            True
9                            True
10                           True
11                           True
12                           True
13                          False
14                          False
15                          False
16                          False
17                          False
18                          False
19                          False
20                          False
21                          False
22                          False
23                          False
24                          False
25                          False
```

[14]:

```
selection_table = neumann["selection_table"]
if HAS_PLOTS:
    fig, ax = plt.subplots(figsize=(9, 4.5), dpi=140)
    sns.lineplot(data=selection_table, x="ii_chain_order",␣
 ↪y="relative_contribution", hue="variant", marker="o", ax=ax)
    sns.lineplot(data=selection_table, x="ii_chain_order", y="spectral_bound",␣
 ↪hue="variant", marker="s", linestyle="--", ax=ax, legend=False)
    ax.set_xlabel("I-I chain order")
    ax.set_ylabel("Relative contribution / spectral bound")
    ax.set_title("Neumann model selection for inhibitory-chain depth")
    plt.tight_layout()
    fig.savefig(OUTPUT_DIR / "neumann_ii_model_selection_by_variant.png",␣
 ↪bbox_inches="tight")
    plt.show()
```

```
else:
    selection_table[["variant", "ii_chain_order", "relative_contribution",␣
 ↪"cumulative_energy", "spectral_bound", "null_z"]]
```



1.7 Who carries stabilizing disinhibition?

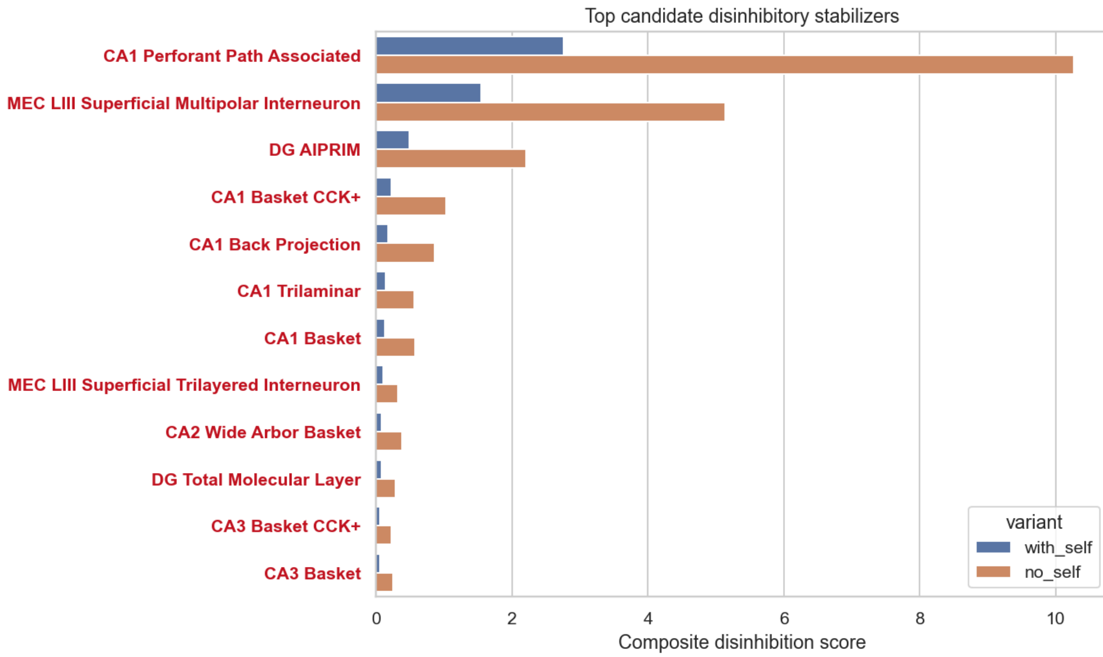
This ranking scores inhibitory cells by their I->E strength, E->I recruitment, and participation in the selected I-I chain order. These are the candidate cell classes to inspect as stabilizing disinhibitory relays.

[15]:
```
disinhibitors = results["disinhibitors"]
display(style_inhibitory_celltypes(disinhibitors))
```

```
<pandas.io.formats.style.Styler at 0x220db49bd90>
```

[16]:
```
if HAS_PLOTS:
    fig, ax = plt.subplots(figsize=(10, 6), dpi=140)
    plot_frame = disinhibitors.groupby("variant", group_keys=False).head(12)
    sns.barplot(data=plot_frame, y="inhibitory_cell", x="score", hue="variant",␣
 ↪ax=ax)
    ax.set_title("Top candidate disinhibitory stabilizers")
    ax.set_xlabel("Composite disinhibition score")
    ax.set_ylabel("")
    color_inhibitory_ticklabels(ax)
    plt.tight_layout()
    fig.savefig(OUTPUT_DIR / "top_disinhibitory_sources_by_variant.png",␣
 ↪bbox_inches="tight")
    plt.show()
else:
```

```
    display(style_inhibitory_celltypes(disinhibitors.groupby("variant",␣
↪group_keys=False).head(12)))
```



1.8 Enumerate E-I-E and E-I-I-E configurations

These tables list the strongest explicit path configurations. `E-I-E` is a two-hop inhibitory route. `E-I-I-E` is the shortest disinhibitory route, with two inhibitory edges producing a positive E-to-E contribution.

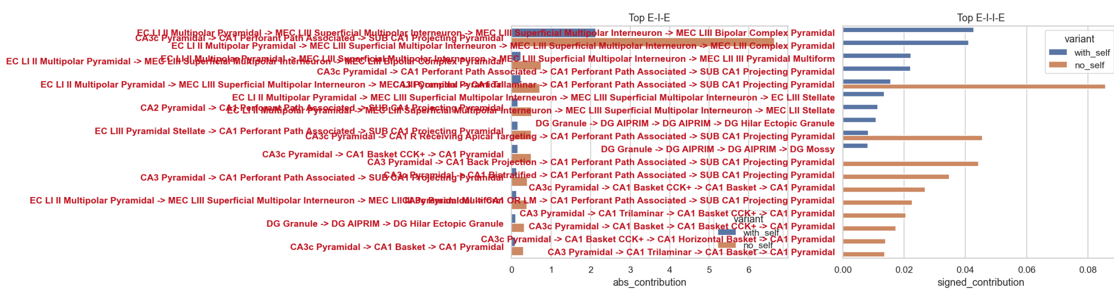
[17]:
```
paths = results["paths"]
print("Top E-I-E configurations")
display(style_inhibitory_celltypes(paths["EIE"]))
print("Top E-I-I-E disinhibitory configurations")
display(style_inhibitory_celltypes(paths["EIIE"]))
```

```
Top E-I-E configurations

<pandas.io.formats.style.Styler at 0x220dcb7bbd0>

Top E-I-I-E disinhibitory configurations

<pandas.io.formats.style.Styler at 0x220dcbd3e10>
```

[18]:
```
if HAS_PLOTS:
    fig, axes = plt.subplots(1, 2, figsize=(13, 5), dpi=140)
    for ax, title, frame, xcol in [
        (axes[0], "Top E-I-E", paths["EIE"].groupby("variant",␣
↪group_keys=False).head(10), "abs_contribution"),
```

```
        (axes[1], "Top E-I-I-E", paths["EIIE"].groupby("variant",␣
↪group_keys=False).head(10), "signed_contribution"),
    ]:
        plot_frame = frame.copy()
        plot_frame["configuration"] = plot_frame.apply(
            lambda r: f"{r['E_source']} -> {r.get('I_middle', r.
↪get('I_first'))} -> {r.get('I_second', '')} -> {r['E_receiver']}".replace("␣
↪->  -> ", " -> "),
            axis=1,
        )
        sns.barplot(data=plot_frame, y="configuration", x=xcol, hue="variant",␣
↪ax=ax)
        ax.set_title(title)
        ax.set_ylabel("")
        color_inhibitory_ticklabels(ax)
    plt.tight_layout()
    fig.savefig(OUTPUT_DIR / "top_path_configurations_by_variant.png",␣
↪bbox_inches="tight")
    plt.show()
else:
    display(style_inhibitory_celltypes(paths["EIE"].head(10)))
```

```
C:\Users\ranic\AppData\Local\Temp\ipykernel_7464\1783642997.py:16: UserWarning:
Tight layout not applied. tight_layout cannot make Axes width small enough to
accommodate all Axes decorations
  plt.tight_layout()
```



1.9 Save tables

[19]:
```
save_side_by_side_outputs(OUTPUT_DIR, results)
print(f"Saved side-by-side tables and figures to {OUTPUT_DIR.resolve()}")
```

```
Saved side-by-side tables and figures to
C:\Users\ranic\Documents\GitHub\schur_decomp\outputs\inhibitory_schur_modulation
```